Keaton Van Patten

AI Engineer · Full-Stack LLM Systems

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SUMMARY

AI engineer who ships production LLM systems end-to-end. Solo-built a 200+ endpoint AI companion platform (FastAPI + Svelte 5 + Electron), then decomposed it into 4 services coordinated over an MQTT message bus — tiered multi-model routing across 4 providers, agent orchestration, and real-time streaming. 5+ years running internal software and IT for a mortgage firm. Looking to build production AI tools and internal platforms.

TECHNICAL SKILLS

Languages Python, JavaScript / TypeScript, SQL, HTML/CSS, PowerShell

Backend & Infra FastAPI, REST APIs, WebSockets / SSE, OAuth / JWT, PostgreSQL + Alembic, Microsoft Graph API, Microsoft Entra ID (Azure AD), Docker, Git, Tailscale

Frontend & Desktop Svelte 5 (runes), React 18, Vite, Tailwind CSS, Electron, Capacitor (mobile / PWA)

AI / ML LLM integration (Claude, DeepSeek, Ollama, OpenAI), model routing & tiering, agent orchestration, RAG / vector embeddings (LanceDB), Whisper (STT), ElevenLabs (TTS), Stable Diffusion (DirectML)

PROJECTS

Companion Hub

AI companion desktop platform · solo · 2025–present

A full-stack AI companion platform with 200+ API endpoints — FastAPI backend, Svelte 5 frontend, Electron shell, packaged for Android via Capacitor.

- Decomposed the monolith into **4 independent apps on an MQTT message bus** (Aedes broker) — per-app FastAPI nodes sharing one engine, with cross-app events and a shared desktop shell.
- Built a **tiered multi-model router** across Claude, DeepSeek, Ollama & OpenRouter that dispatches by task complexity and cost, with automatic local fallback and per-project cost tracking.
- Built an **agent orchestration system**: rule-based dispatch classifier, DeepSeek planning engine, and a background plan executor with phase-by-phase tracking.
- Engineered **real-time token streaming (SSE)**, a **tool-execution engine** (file / git / shell) with path-safety guards, **voice I/O** (Whisper / ElevenLabs), and **local image generation** (Stable Diffusion).

Python 3.12, FastAPI, Svelte 5, Electron 35, PostgreSQL, Ollama, DeepSeek, Capacitor

OpenClaw AI Gateway

Self-hosted multi-provider AI infrastructure · 2025–present

- Self-hosted and operate three isolated gateway instances with **model routing** across DeepSeek, Anthropic (OAuth), Ollama, HuggingFace, Venice AI & OpenRouter.
- Configured **LanceDB + nomic-embed vector memory** for persistent agent context; auto-restart lifecycle and scoped execution-approval tokens over a local socket.

Node.js, WebSocket, OAuth 2.0, LanceDB, SQLite

WorkTeams

Self-hosted Teams alternative · 2026

- Built a **Microsoft Teams-style desktop app** for a Windows Server 2016 environment — installs from USB and runs locally with no internet required.
- **Real-time chat** (socket.io), calls, video & calendar; **Microsoft Graph + MSAL** identity pulling directory profiles, chats & meetings; admin panel + Squirrel installer.

React 18, Electron 33, TypeScript, socket.io, MSAL, Microsoft Graph

EXPERIENCE

Systems Manager & Software Developer — HRBC Inc. / Van Patten Mortgage Group

Sept 2020 – Present

- Sole engineer and IT owner for a mortgage firm for **5+ years** — every internal tool and system designed, built, deployed, and maintained solo.
- Built the firm's internal AI stack: **AI-powered email triage** via Microsoft Graph (MSAL OAuth) with categorization, priority scoring & reply drafts; **email-to-task extraction**; AI project trackers with automated code review.
- Administer the Microsoft 365 / Windows Server environment — accounts, permissions, networking, security — and built a CRM-based client referencing system used daily.

Utility Technician — MP NexLevel

2018 – 2020

EDUCATION & CERTIFICATIONS

Google IT Support Professional Certificate · Google Computer Networking Certificate · Self-taught full-stack & AI/LLM engineering